

Senyi Kong

PhD student in Earth Climate and Environment

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ResearchGate | Expected PhD completion: June 2027

RESEARCH PROFILE

Atmospheric scientist working on electromagnetic scattering by nonspherical and inhomogeneous particles, aerosol and cloud optics, satellite hyperspectral retrieval, and numerical weather prediction data assimilation. Research experience connects physical algorithm development with satellite applications and forecast evaluation.

EDUCATION

PhD in Earth Climate and Environment | Zhejiang University

Sep 2022 - Jun 2027 (expected) | School of Earth Sciences. Direct-entry doctoral programme. All doctoral coursework completed.

BSc in Atmospheric Science | Zhejiang University

Sep 2018 - Jun 2022 | GPA 4.60; ranked first in class.

RESEARCH PROJECT CONTRIBUTIONS

Dates denote project periods; entries describe research contributions within collaborative projects.

All-sky assimilation of FY-4 hyperspectral observations

Jan 2025 - Dec 2028 | National Natural Science Foundation of China (NSFC), Joint Fund

Research on assimilation methods for FY-4 infrared hyperspectral satellite observations.

Electromagnetic scattering by large nonspherical particles

Jul 2023 - Jun 2026 | Chinese Academy of Sciences, major research infrastructure open-access project

Numerical computation of electromagnetic scattering by large atmospheric particles.

All-weather Fengyun satellite data assimilation

Nov 2022 - Oct 2025 | National Key Research and Development Program of China

All-sky assimilation of geostationary hyperspectral infrared data within a global/regional assimilation framework.

Fast scattering calculations for cloud-affected FY-4 observations

Aug 2022 - May 2025 | CMA Earth System Modeling and Prediction Centre, collaborative project

Fast scattering models for nonspherical cloud particles in infrared hyperspectral applications.

FY-4 data assimilation for high-impact weather forecasting

Jul 2023 - Dec 2023 | Zhejiang Institute of Meteorological Sciences, collaborative project

Atmospheric profile retrieval and data assimilation for CMA-MESO.

Scattering by nonspherical, inhomogeneous atmospheric particles

Jan 2021 - Dec 2023 | NSFC, Excellent Young Scientists Fund project

Dust-particle modelling and optical-property calculations.

METHODS AND ACADEMIC COMMUNICATION

Methods: IITM; nonspherical particle modelling; optical-property calculations; radiative transfer; atmospheric profile retrieval; 4DVar data assimilation; Bayesian and machine-learning methods.

English: CET-6 score 592; academic reading, writing, and communication. **Poster:** FY-4A retrieval of dust complex refractive index and mineral composition.

Collaboration and service: Peer support within the research group and volunteer support for student activities. **Interests:** Regular exercise, music, art exhibitions, and live performances.

PUBLICATIONS

1. X. Song, W. Han, **S. Kong**, J. Yan, K. Zhou (2026). Reconstruction of Three-Dimensional Cloud Structures Using FY-4B AGRI Observations and CMA-MESO Forecasts. *Weather and Forecasting*, Early online release, 14 August 2026. 10.1175/WAF-D-25-0251.1
2. L. Bi, **S. Kong**, Y. Xi, X. Wang, W. Lin, Z. Du, W. Han, X. Zhang (2026). ZJU-SCATT-V1.0: A Comprehensive Single-Scattering Database of Nonspherical Aerosol and Cloud Particles. *Bulletin of the American Meteorological Society*, 107(7), E1563-E1578. 10.1175/BAMS-D-25-0258.1
3. **S. Kong**, W. Han, L. Bi, R. Yin (2026). All-sky FY4B-GIIRS infrared hyperspectral satellite data assimilation in the CMA-GFS 4DVar system. Part I: Implementation. *Quarterly Journal of the Royal Meteorological Society*, 152(778), e70185. 10.1002/qj.70185
4. **S. Kong**, W. Han, L. Bi, R. Yin (2026). All-sky FY4B-GIIRS infrared hyperspectral satellite data assimilation in the CMA-GFS 4DVar system. Part II: Evaluation. *Quarterly Journal of the Royal Meteorological Society*, 152(778), e70184. 10.1002/qj.70184
5. **S. Kong**, L. Bi, W. Han, R. Yin, H. Zhang (2026). Hybrid Bayesian-Machine Learning Framework for Multi-Profile Atmospheric Retrieval from Hyperspectral Infrared Observations. *Advances in Atmospheric Sciences*, 43(2), 373-389; online November 2025. 10.1007/s00376-025-5070-9
6. R. Yin, W. Han, Z. Zhao, X. Gong, **S. Kong**, F. Zhang, J. Li (2025). The Preliminary Assimilation and Impacts of FY-4B Geostationary Interferometric Infrared Sounder (GIIRS) Cloud-Cleared Radiances. *Journal of Geophysical Research: Atmospheres*, 130(23), e2025JD043520. 10.1029/2025JD043520
7. **S. Kong**, Z. Wang, L. Bi (2024). Uncertainties in laboratory-measured shortwave refractive indices of mineral dust aerosols and derived optical properties: a theoretical assessment. *Atmospheric Chemistry and Physics*, 24, 6911-6935. 10.5194/acp-24-6911-2024
8. Z. Wang, L. Bi, **S. Kong** (2023). Flexible implementation of the particle shape and internal inhomogeneity in the invariant imbedding T-matrix method. *Optics Express*, 31(18), 29427-29439. 10.1364/OE.498190
9. **S. Kong**, K. Sato, L. Bi (2022). Lidar Ratio-Depolarization Ratio Relations of Atmospheric Dust Aerosols: The Super-Spheroid Model and High Spectral Resolution Lidar Observations. *Journal of Geophysical Research: Atmospheres*, 127(4), e2021JD035629. 10.1029/2021JD035629

PATENT

An iterative bias-correction method accounting for non-uniformity among satellite instrument detector elements.
Second-listed inventor. Application CN202511848925.6; filed December 2025. Granted.

HONOURS AND AWARDS

2025 | Labour Education Role Model, Zhejiang University
2024; 2023 | Outstanding Graduate Student, Zhejiang University
2023 | Graduate Student Award for All-Round Excellence, Zhejiang University
2023 | Top Ten Doctoral Research Advances, School of Earth Sciences, Zhejiang University
2022 | Top Ten Poster Award, School of Earth Sciences, Zhejiang University
2018-2022 | National Scholarship; First-Class Scholarship, Zhejiang University
2018-2022 | Outstanding Graduate, Zhejiang Province; Outstanding Youth League Cadre and Member, Zhejiang University
2019-2021 | First Prize, 11th National Undergraduate Mathematics Competition (non-mathematics category); Second Prize, Zhejiang Undergraduate Physics Innovation Competition (theory)